

You Suck at Prompting (But Here's How to Fix It)

1. Executive Summary

The content argues that poor results from Generative AI are typically a "skill issue" on the part of the user rather than a failure of the technology. Prompting is defined not as asking questions, but as programming an LLM (Large Language Model) to complete a pattern. The video outlines a progression of techniques to improve output quality, starting with foundational concepts like Personas and Context, moving to formatting, and culminating in advanced strategies like Chain of Thought, Tree of Thoughts, and Adversarial Validation. The ultimate "meta-skill" required for effective prompting is clarity of thought—the ability to clearly describe a system or problem before engaging the AI.

2. Core Knowledge & Concepts

The Nature of Prompting

- **Definition:** According to Dr. Jules White (Vanderbilt University), a prompt is a "call to action" to the Large Language Model. It is not just a question; it is a program written in words.
- **Mechanism:** LLMs are "prediction engines" or advanced autocomplete systems. They do not "think" in the human sense; they predict the next statistically probable word based on the pattern provided.
- **The Pattern Principle:** If a prompt (pattern) is vague, the AI guesses based on general probability. If the pattern is focused, the AI's prediction becomes more accurate.

Foundational Techniques

- **Personas:**
 - **Concept:** Assigning a specific role or identity to the AI (e.g., "Senior Site Reliability Engineer").
 - **Function:** This narrows the focus of the expertise the AI draws from, moving it from a generic "nobody" to a specific expert perspective.
 - **System vs. User Prompts:** In API or advanced interfaces (like Claude Code), personas are often placed in the "System Prompt" to instruct the AI on who it is, while the specific task goes in the "User Prompt."
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- **Context:**

- **Importance:** Described as the most important technique. "Context is King."
- **Function:** Providing specific facts and details prevents the AI from "hallucinating" (making things up to fill gaps).
- **Hallucination Fix:** Explicitly giving the AI permission to fail or say "I don't know" if the answer isn't in the provided context reduces fabrications.
- **Memory Warning:** While some LLMs have memory features, users should never assume the AI remembers previous context. "Always provide all the context every time."
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- **Format:**
 - **Concept:** Telling the LLM exactly how the result should look (e.g., bullet points, tone, word count).
 - **Benefit:** Standardizes the output and prevents generic, "corporate fluff" responses.
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Advanced Techniques

- **Chain of Thought (COT):**
 - **Concept:** Instructing the AI to "show its work" or think step-by-step before answering.
 - **Benefits:** Increases accuracy (the AI "thinks" before writing) and increases user trust (the user can verify the logic).
 - **Implementation:** Many models now have this baked in (e.g., "Extended Thinking" or "Reasoning Models").
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- **Few-Shot Prompting:**
 - **Concept:** Providing examples of "good" output within the prompt itself.
 - **Function:** Instead of describing the output, the user *shows* the output pattern they want the AI to repeat.
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- **Tree of Thoughts (TOT):**
 - **Concept:** Asking the AI to explore multiple paths or branches of reasoning simultaneously, rather than a single linear path.
 - **Benefit:** Allows for self-correction. If one path is a dead end, the AI can switch to a better one, generating a diversity of options for complex problem-solving.
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- **The Playoff Method (Adversarial Validation):**
 - **Concept:** Simulating a competition or "Battle of the Bots."
 - **Mechanism:** Creating distinct personas (e.g., an Engineer vs. an Angry Customer) to critique each other's work and synthesize a final version based on that feedback.
 - **Why it works:** AI is often better at critiquing and editing than original writing.
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The Meta-Skill

- **Clarity of Thought:** The underlying skill required for all prompting. If a user cannot explain their goal clearly to themselves (e.g., on a notepad), they cannot prompt the AI effectively.
- **Workflow:** Think first, prompt second. Red-team your own ideas to ensure robustness before inputting them into the AI.

3. Verbatim Examples & Scripts

Context: Demonstration of how LLMs act as prediction engines/autocomplete using a specific phrase.

Content:

"The early bird gets the **worm**"

Context: Testing Google Gemini's ability to predict a specific catchphrase based on vague vs. specific patterns.

Content:

Vague Prompt: "You need to learn docker"

Result: "You need to learn Docker to easily build, deploy, and run applications..."

Specific Prompt: "You need to learn docker _____!!!!"

Result: "You need to learn Docker **RIGHT NOW!!!!**"

Context: The "Persona" prompt used to improve a generic apology email.

Content:

"Act as a Senior Site Reliability Engineer at Cloudflare writing to both enterprise customers and fellow engineers. Write an apology email for today's 6-hour outage that affected major services."

Context: The "Context" prompt, providing specific facts to prevent hallucination.

Content:

"You are a Senior Site Reliability Engineer at Cloudflare writing to both enterprise customers and fellow engineers. Write an apology email for today's 6-hour outage that affected major services."

Here are the FACTS about today's outage. *IMPORTANT! ALSO perform a web search to learn about this outage and EXACTLY what happened.

INCIDENT DETAILS:

- We made a database permissions change at 02:47 UTC
- This caused duplicate metadata entries in our Bot Management feature configuration file

- The Bot Management feature file doubled in size (from ~60 features to 200+ features)
- Our system has a hardcoded 200-feature limit that was never documented - When the file exceeded 200 features, the Bot Management system crashed
- This cascaded to affect CDN traffic routing - Major sites went down for 6 hours: X, ChatGPT, Spotify, Discord, and others
- We fixed it by reverting the permissions change and removing duplicate entries

Write an apology email explaining what happened."

Context: The "Format" prompt, specifying output requirements to remove corporate fluff.

Content:

"OUTPUT REQUIREMENTS:

- 1 - Format: Use a clear bulleted list for the timeline of events.
- 2 - Length: Keep it under 200 words.
- 3 - Tone: Professional, apologetic, and radically transparent. No corporate fluff."

Context: An "off the rails" Format prompt example to demonstrate control over tone.

Content:

"OUTPUT REQUIREMENTS:

- 1 - Tone: Extremely anxious and panicked. Sound like you are afraid of getting fired.
- 2 - Length: Keep it under 200 words.
- 3 - Format: Run-on sentences, all lowercase."

Context: The "Chain of Thought" prompt instruction.

Content:

"Before writing the email, think through step-by-step:

1. What was the root cause? (The permissions change)
2. Why did a permissions change create duplicate metadata? (Explain the mechanism)
3. Why did duplicate entries cause the file size to exceed 200? (Size doubling)
4. Why did hitting the 200-feature limit crash the system? (Undocumented hardcoded limit)
5. How did a bot management crashing affect CDN routing? (Cascade explanation)
6. How did we fix it? (Revert + manual cleanup)
7. What are we doing to prevent this? (Future safeguards)

Then write the email explaining this clearly for both technical and non-technical audiences, using 'BUT/THEREFORE' transitions to show causal flow."

Context: The "Tree of Thoughts" prompt to brainstorm strategic approaches.

Content:

"You are a Senior Site Reliability Engineer (SRE) at Cloudflare with extensive experience in incident communication and post-mortem writing.

We need to draft a customer-facing apology email following a significant outage. Please use the 'Tree of Thoughts' reasoning framework to determine the most effective communication strategy. Structure your response exactly as follows:

Step 1: Brainstorm three distinct tonal/strategic approaches (branches) for the apology email:

- Branch A: 'Radical Transparency' — Heavy emphasis on the detailed technical root cause and timeline.
- Branch B: 'Customer Empathy First' — Primary focus on acknowledging the business impact, frustration, and erosion of trust.
- Branch C: 'Future-Focused Assurance' — Center the message almost entirely on what has been fixed and the new safeguards implemented.

Step 2: Evaluate each branch objectively:

- List clear pros and cons for each approach.
- Specifically analyze: Which restores trust most effectively? Which risks alienating or confusing non-technical customers? Which might invite unwanted scrutiny or speculation?

Step 3: Synthesize the best elements from all three branches to create an optimal 'Golden Path' strategy that balances transparency, empathy, and confidence-building.

Step 4: Using the Golden Path, write the complete customer apology email."

Context: The "Playoff Method" (Adversarial Validation) prompt.

Content:

"You are the 'Prompt Engineering Tournament Controller.'

Your task is to generate the perfect public apology email from Cloudflare following a major outage.

To achieve this, simulate a 3-round competition between three distinct personas:

Contestants

- Contestant 1 - The Engineer: Obsessed with technical accuracy, transparency, and factual precision. Hates Fluff and vagueness.
- Contestant 2 - The PR Crisis Manager: Focused on protecting brand reputation, maintaining customer trust, and managing sentiment. Skilled at empathetic language and damage control.

- Contestant 3 - The Angry Customer: A CTO whose company just lost \$50,000+ due to the outage. Furious, zero tolerance for excuses, calls out corporate speak instantly.

Tournament Structure (3 Rounds)

Round 1: Initial Drafts

- The Engineer writes their version of the apology email.
- The PR Crisis Manager writes their version of the apology email.
(Do not show the Angry Customer yet.)

Round 2: The Roast

- The Angry Customer reads both drafts and brutally critiques them.
- They must point out every excuse, every vague or evasive sentence, every tone-deaf moment, and anything that feels insincere or inadequate.

Round 3: The Synthesis

- The Engineer and PR Crisis Manager read the Angry Customer's feedback.
- They collaborate (you show their brief discussion if helpful) and produce ONE final, unified apology email that fully addresses every single criticism raised in Round 2."

4. Key Data Points

- **Ethan Mollick (Professor at Wharton University):** Stated that "95% of all practical problems folks encounter can be solved by turning on Extended Thinking."
- **Joseph Thacker (The Promptfather):** Advises treating bad AI responses as a "personal skill issue." If the response is bad, it means the user didn't explain it well enough or provide enough context.
- **Eric Pope (NetworkChuck Academy):** "The more specific you could get at later stages, the better results you'll get."
- **Daniel Miessler (Creator of Fabric):** His "pinnacle superpower" is not tech stuff, but "clear thinking." He sits down and describes exactly how he wants something to work before attempting to convert it into tech/prompts.